

Task API – CRUD To-Do List

Overview

Task API is a RESTful CRUD (Create, Read, Update, Delete) application built with **Python**, **FastAPI**, and **Uvicorn**. The API allows users to create, retrieve, update, and delete tasks stored in an in-memory list.

This project was developed as part of the **Build Your First CRUD API** assignment to demonstrate the fundamentals of REST API development, HTTP methods, status codes, request validation, Swagger documentation, and GitHub version control.

GitHub Repository: [urlFlyRankWeek1-Build-your-first-CRUD-APIhttps://github.com/amuk86/FlyRankWeek1-Build-your-first-CRUD-API](https://github.com/amuk86/FlyRankWeek1-Build-your-first-CRUD-API)

Features

- Create new tasks
 - Retrieve all tasks
 - Retrieve a task by its ID
 - Update existing tasks
 - Delete tasks
 - Input validation
 - Correct HTTP status codes
 - Interactive Swagger UI documentation
 - In-memory data storage
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Technologies Used

- Python 3.10+
 - FastAPI
 - Uvicorn
 - Pydantic
 - Swagger UI (built into FastAPI)
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Project Structure

```
task-api/  
|— main.py
```

```
|— requirements.txt
|— README.md
└─ screenshots/
    └─ swagger-ui.png
```

Installation

Clone the Repository

```
git clone https://github.com/amuk86/FlyRankWeek1-Build-your-first-CRUD-API.git
cd FlyRankWeek1-Build-your-first-CRUD-API
```

Install Dependencies

```
pip install -r requirements.txt
```

Running the Application

Start the development server with:

```
uvicorn main:app --reload
```

The API will be available at:

```
http://localhost:8000
```

Swagger documentation is available at:

```
http://localhost:8000/docs
```

ReDoc documentation is available at:

```
http://localhost:8000/redoc
```

API Endpoints

Method	Endpoint	Description
GET	/	Returns API information
GET	/health	Checks whether the API is running
GET	/tasks	Retrieves all tasks
GET	/tasks/{id}	Retrieves a task by ID
POST	/tasks	Creates a new task
PUT	/tasks/{id}	Updates an existing task
DELETE	/tasks/{id}	Deletes a task

Example Request

Create a new task:

```
curl -i -X POST http://localhost:8000/tasks
-H "Content-Type: application/json"
-d '{"title":"Buy milk"}'
```

Example response:

```
HTTP/1.1 201 Created

{
  "id": 4,
  "title": "Buy milk",
  "done": false
}
```

HTTP Status Codes

Status Code	Description
200	Request completed successfully

Status Code	Description
201	Resource created successfully
204	Resource deleted successfully
400	Invalid request
404	Resource not found

Validation

The API validates all incoming requests.

Validation rules:

- The `title` field is required.
- The title cannot be empty.
- Invalid requests return a **400 Bad Request** response with an error message.

Swagger Documentation

FastAPI automatically generates interactive API documentation using Swagger UI.

Open the following URL after starting the server:

```
http://localhost:8000/docs
```

Add a screenshot of your Swagger UI in the repository once you have completed testing.

Testing

The API was tested using:

- Browser
- curl
- Swagger UI

The following CRUD operations were successfully tested:

- Create
 - Read
 - Update
 - Delete
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In-Memory Storage

This project stores task data in memory.

This means:

- Tasks are stored only while the server is running.
 - Restarting the server resets the task list to the initial sample data.
 - No database or file storage is used.
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GitHub Commits

The project was developed using incremental commits for each assignment stage:

- Stage 0 – Hello Server
 - Stage 1 – Root and Health Endpoints
 - Stage 2 – Read Endpoints
 - Stage 3 – Create Endpoint
 - Stage 4 – Full CRUD
 - Stage 5 – Swagger UI
 - Stage 6 – Publish to GitHub
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Future Improvements

Possible enhancements include:

- Database integration
- User authentication
- Task filtering
- Search functionality
- Pagination
- Statistics endpoint
- Persistent data storage

Author

Amukelani Mabunda

GitHub:  github.com/amuk86  <https://github.com/amuk86> 

License

This project was developed for educational purposes as part of the FlyRank Backend Development program.